MCA 1st Semester Examination, 2023 (R & B)

Subject : MCA

Paper –MCC-1.1.1

Full Marks : 80

Time : 3 hours

Answer **all** questions

The figures in the right-hand margin indicate marks

Candidates are required to give their answers in their own words as far as practicable

UNIT–I

1. (*a*) State and prove Pigeonhole principle. 10

(*b*) Prove that two integer a and b have the same remainder when divided by a same integer n iff the integer $a-b$ is divisible by n. 10

(*Turn Over*)

Or

(*c*) Prove or disprove the validity of the following argument

"Every living thing is a plant or an animal.

David's dog is alive and it is not a plant.

All animals have hearts.

Hence David's dog has a heart. 10

(*d*) Find and prove a formula for the sum of first n cubes. 10

UNIT—II

2. (*a*) Write special properties of a binary relation and restate those in terms of diagraphs. 10

(*b*) Define a poset, Lattice, maximal and minimal elements, greatest and least elements of a poset. Use the poset $[A; 1]$ where $A = \{2, 3, 4, 6, 8, 24, 48\}$ to explain the above. 10

Or

(c) R be a relation on $A = \{a, b, c, d\}$ whose adjacency matrix is

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Compute the adjacency matrix of R^+ using Warshall's algorithm. 10

(d) Solve $a_n - 5a_{n-1} + 6a_{n-2} = 2^n$. 10

UNIT—III

3. (a) (i) If $V = \{v_1, \ldots, v_n\}$ be a vertex set of a non-directed graph G, then prove that 2

$$\sum_{i=1}^{n} \deg(v_i) = 2|E|$$

(*ii*) In a non-directed graph prove that there is an even number of vertices of odd degree. 8

(*b*) (*i*) Write the expression

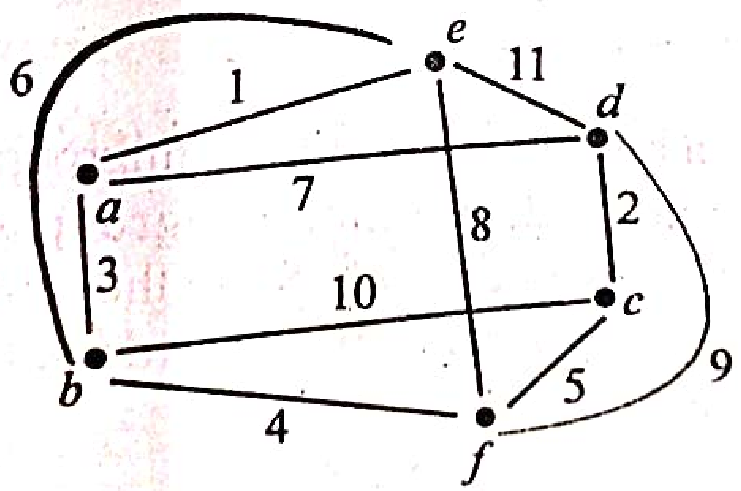
$$\{[(a+b)\times c]\times(d+e)\}-[f-(g\times h)]$$

as a tree. 5

(*ii*) Prove that a tree of n-vertices has exactly $n-1$ edges. 5

Or

(*c*) Define spanning tree of a graph. Write Kruskal's algorithm for finding minimal spanning tree. Using this obtain minimal spanning tree of the graph 10



(*d*) Prove the following :

(*i*) A complete graph K_n is planar iff $n \leq 4$. 5

(*ii*) A complete bipartite graph Km, n is planar iff $m \leq 2$. 5

UNIT–IV

4. (*a*) Define Boolean Algebra. Show that a Boolean algebra is associative under operations of + and . . 10

(*b*) Obtain sum-of-product canonical form of the Boolean expression

$$\bar{x}_1 + [(x_2 + \bar{x}_3)(\overline{x_2 x_3})](x_1 + x_2 \bar{x}_x)$$ 10

Or

(*c*) Define a binary operation, a semigroup, a cyclic monoid, normal subgroup and give examples of each. State the difference between a ring and a field with an example. 10

(*d*) Define maxterm and minterm in a Boolean algebra. If minterm form of

$$f(x,y)=(\alpha\wedge\bar{x}\wedge\bar{y})\vee(\beta\wedge\bar{x}\wedge y)$$
$$\vee(0\wedge x\wedge\bar{y})\vee(1\wedge x\wedge y),$$

give the maxterm normal form of f. 10
